

Practice Test III (Version 1)

Name
Date
Course

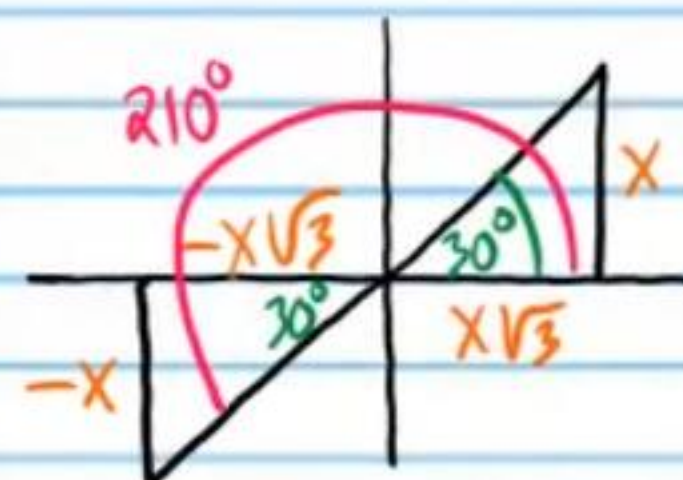
- I) Write your name, the date, and the course title (i.e. Trigonometry).
- II) Show all work.
- III) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) You will need a calculator for this test (be sure that it is in degree mode).
- V) Each problem is worth four points for a total of 100 points.

Find the values of the following expressions without a calculator.

① $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

$$\theta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$\tan \theta = \frac{1}{\sqrt{3}}$$



$$\theta = 30^\circ$$

$$\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \boxed{30^\circ}$$

Find the value of the following expression. You will need a calculator. Round to the nearest tenth.

③ $\sin^{-1}\left(\frac{2}{3}\right) \approx \boxed{41.8^\circ}$

② $\sec^{-1}(-1)$

$$\theta = \sec^{-1}(-1)$$

$$\sec \theta = -1$$

$$\theta = 180^\circ$$

$$\sec^{-1}(-1) = \boxed{180^\circ}$$

Write an equation corresponding to each graph.
★ There is more than one equation that can be written for each graph (e.g. cosine functions or sine functions can represent the same graph). To avoid confusion, use the sine or cosecant functions rather than the cosine or secant functions. ★ Each horizontal unit on the graphs is 90° . Each vertical unit is one.

④ General Form of the Sine Function: $y = A \sin(\omega x - \phi) + B$

Midline Equation: $y = -1$

$$B = -1$$

Amplitude: $A = 2$

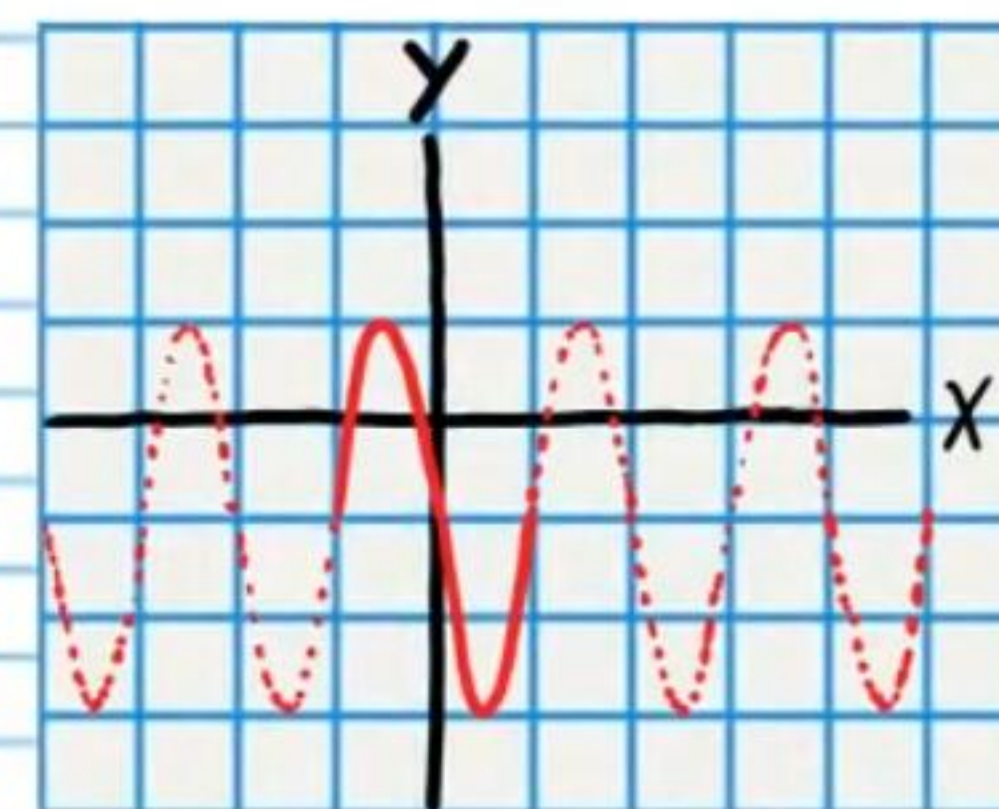
Omega: $P = \frac{360^\circ}{\omega}$

$$180^\circ = \frac{360^\circ}{\omega}$$

$$\omega \cdot 180^\circ = \frac{360^\circ}{\omega} \cdot \omega$$

$$180^\circ \omega = 360^\circ$$

$$\omega = 2$$



Phase Shift:

$$PS = \frac{\phi}{\omega}$$

$$-90^\circ = \frac{\phi}{2}$$

$$-180^\circ = \phi$$

$$y = 2 \sin(2x - (-180^\circ)) + (-1)$$

$$\boxed{y = 2 \sin(2x + 180^\circ) - 1}$$

⑤ General Form of the Cosecant Function: $y = A \csc(\omega x - \Phi) + B$

Midline Equation: $y = -1$

$$B = -1$$

A: $A = 1$

Omega: $P = \frac{360^\circ}{\omega}$

$$360^\circ = \frac{360^\circ}{\omega}$$

$$\omega = 1$$

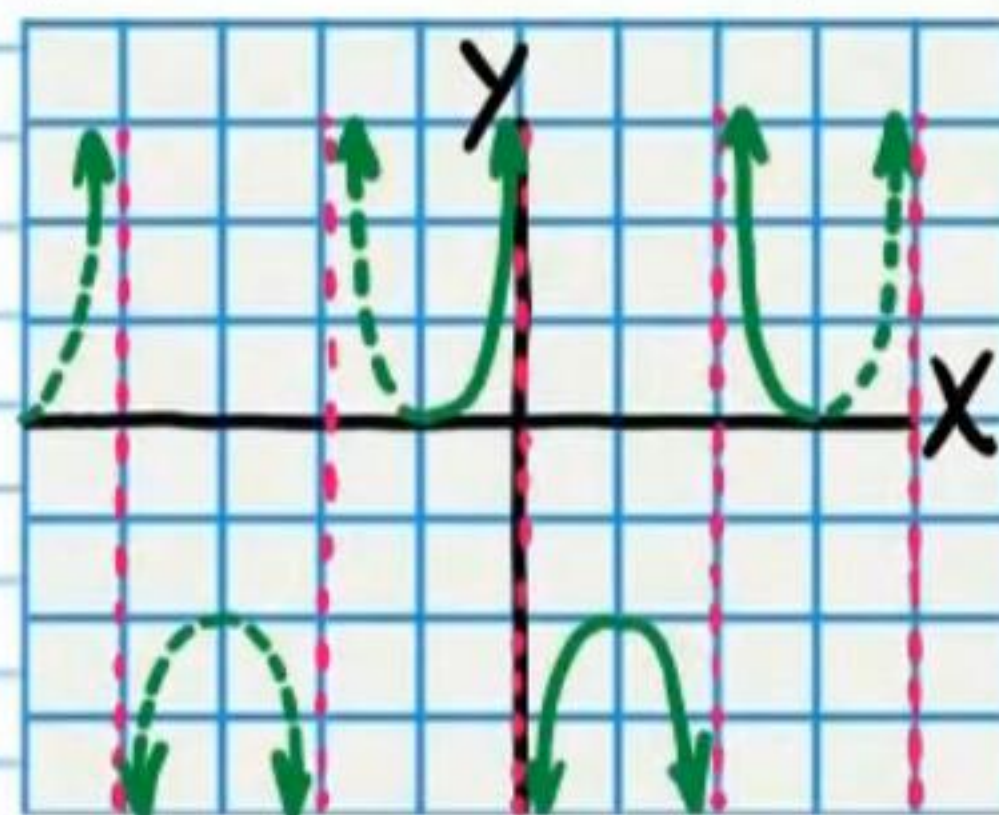
Phase Shift: $PS = \frac{\Phi}{\omega}$

$$-180^\circ = \frac{\Phi}{1}$$

$$\Phi = -180^\circ$$

$$y = (1) \csc((1)x - (-180^\circ)) + (-1)$$

$$y = \csc(x + 180^\circ) - 1$$



⑥ General Form of the tangent Function: $y = A \tan(\omega x - \Phi) + B$

Midline Equation: $y = 0$

$$B = 0$$

A: ~~Assume~~ $A = 1$

Omega: $P = \frac{180^\circ}{\omega}$

$$360^\circ = \frac{180^\circ}{\omega}$$

$$\omega = \frac{1}{2}$$

Phase Shift: $PS = \frac{\Phi}{\omega}$

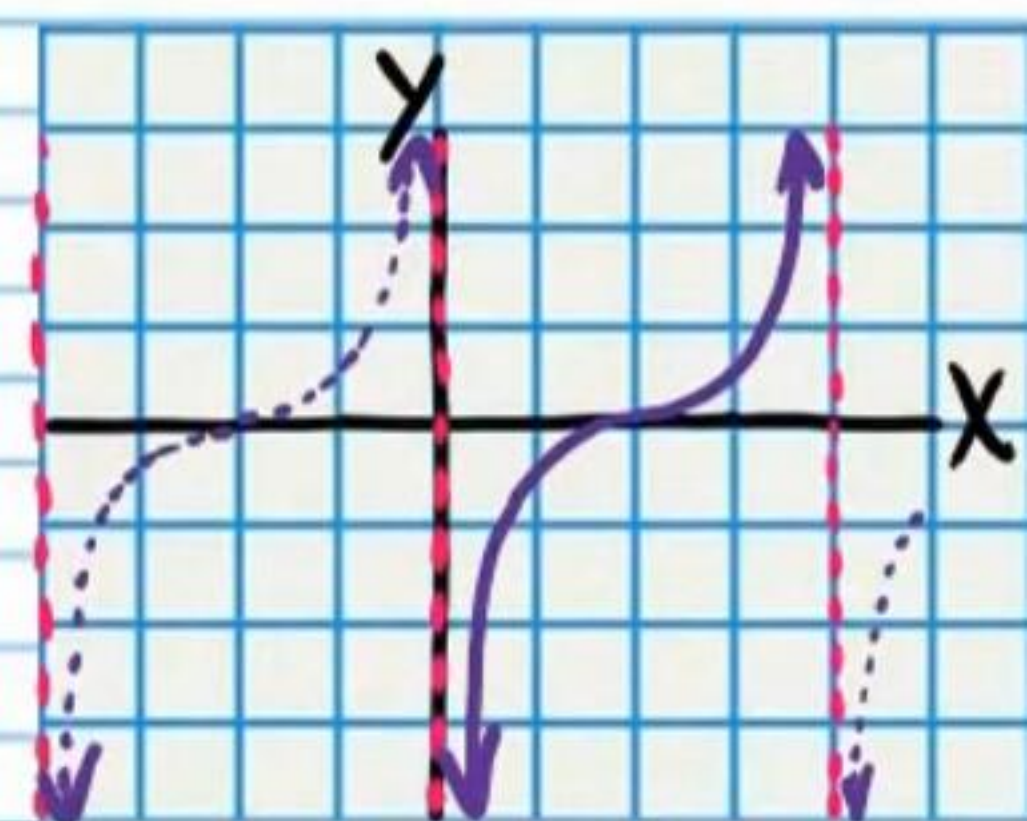
$$180^\circ = \frac{\Phi}{\frac{1}{2}}$$

$$\left(\frac{1}{2}\right) 180^\circ = \frac{\Phi}{\frac{1}{2}} \left(\frac{1}{2}\right)$$

$$90^\circ = \Phi$$

$$y = (1) \tan\left(\frac{1}{2}x - 90^\circ\right) + 0$$

$$y = \tan\left(\frac{x}{2} - 90^\circ\right)$$



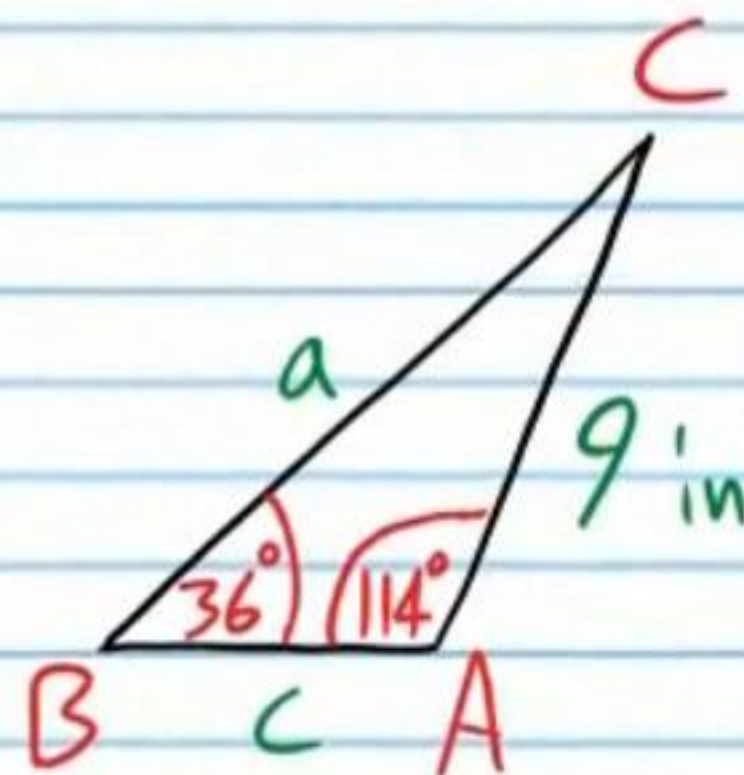
Find the missing side lengths and angle measures of each triangle. In other words, solve each triangle. There might not be a triangle that exists for some of these problems.

⑦ $\triangle C$

$$A + B + C = 180^\circ$$

$$114^\circ + 36^\circ + C = 180^\circ$$

$$\boxed{C = 30^\circ}$$



⑧ a

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin 114^\circ}{a} = \frac{\sin 36^\circ}{9}$$

$$9 \sin 114^\circ = a \sin 36^\circ$$

$$\frac{9 \sin 114^\circ}{\sin 36^\circ} = \frac{a \sin 36^\circ}{\sin 36^\circ}$$

$$a = \frac{9 \sin 114^\circ}{\sin 36^\circ}$$

$$\boxed{a \approx 13.9879 \text{ in}}$$

⑨ c

$$\frac{\sin C}{c} = \frac{\sin B}{b}$$

$$\frac{\sin 30^\circ}{c} = \frac{\sin 36^\circ}{9}$$

$$c = \frac{9 \sin 30^\circ}{\sin 36^\circ}$$

$$\boxed{c \approx 7.6559 \text{ in}}$$

⑩ $\triangle B$

$$A + B + C = 180^\circ$$

$$87^\circ + B + 70^\circ = 180^\circ$$

$$\boxed{B = 23^\circ}$$

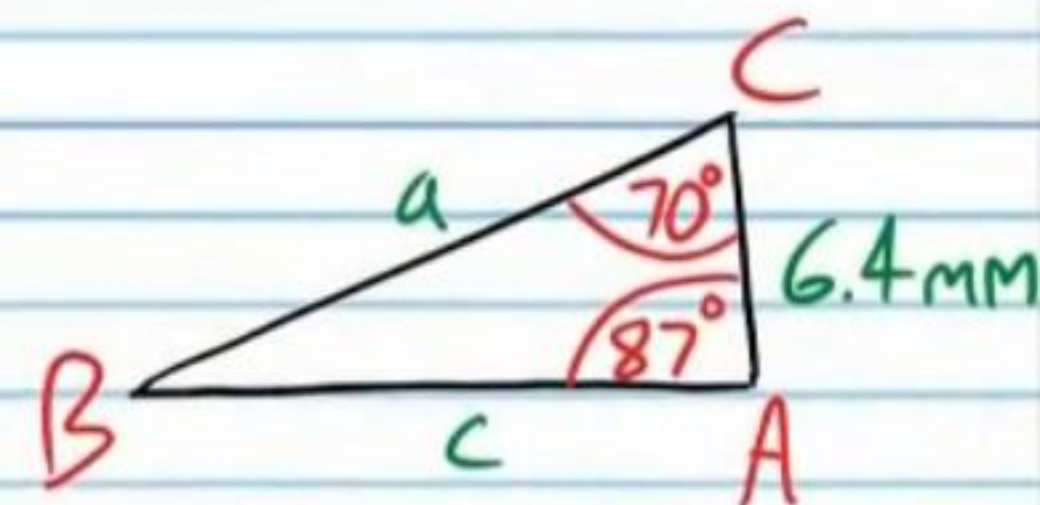
⑪ a

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin 87^\circ}{a} = \frac{\sin 23^\circ}{6.4}$$

$$a = \frac{6.4 \sin 87^\circ}{\sin 23^\circ}$$

$$\boxed{a \approx 16.3571 \text{ mm}}$$



⑫ c

$$\frac{\sin C}{c} = \frac{\sin B}{b}$$

$$\frac{\sin 70^\circ}{c} = \frac{\sin 23^\circ}{6.4}$$

$$c = \frac{6.4 \sin 70^\circ}{\sin 23^\circ}$$

$$\boxed{c \approx 15.3917 \text{ mm}}$$

⑬ $m \angle B$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$20^2 = 10^2 + 13^2 - 2(10)(13) \cos B$$

$$400 = 100 + 169 - 260 \cos B$$

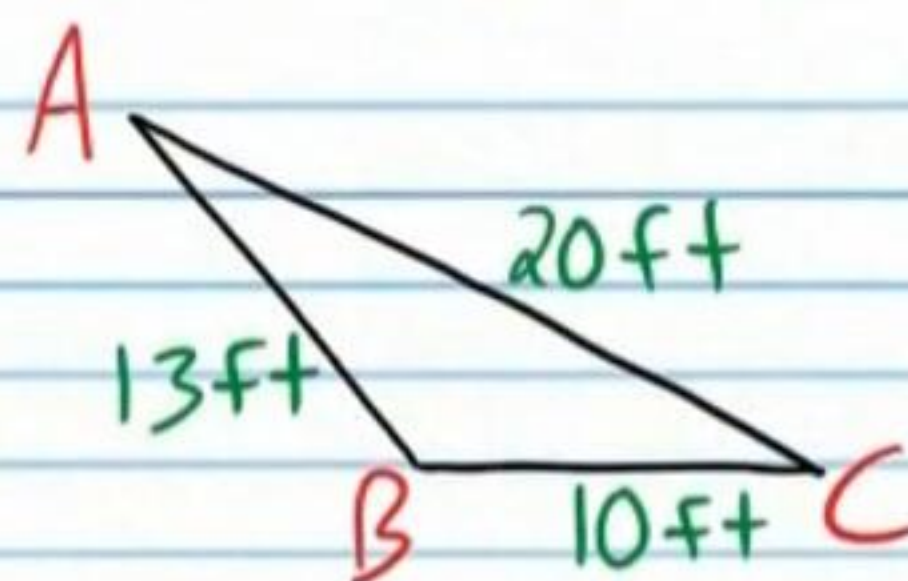
$$400 = 269 - 260 \cos B$$

$$131 = -260 \cos B$$

$$-0.503846... \approx \cos B$$

$$B \approx \cos^{-1}(-0.503846...)$$

$$\boxed{B \approx 120.25479^\circ}$$



⑭ $m \angle A$

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin A}{10} \approx \frac{\sin 120.25479^\circ}{20}$$

$$\sin A \approx \frac{10 \sin 120.25479^\circ}{20}$$

$$\sin A \approx 0.43189...$$

$$A \approx \sin^{-1}(0.43189...)$$

$$\boxed{A \approx 25.58799^\circ}$$

⑮ $m \angle C$

$$A + B + C = 180^\circ$$

$$25.58799^\circ + 120.25479^\circ + C \approx 180^\circ$$

$$\boxed{C \approx 34.157^\circ}$$

⑯ c

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$c^2 = 8.8^2 + 11^2 - 2(8.8)(11) \cos 128^\circ$$

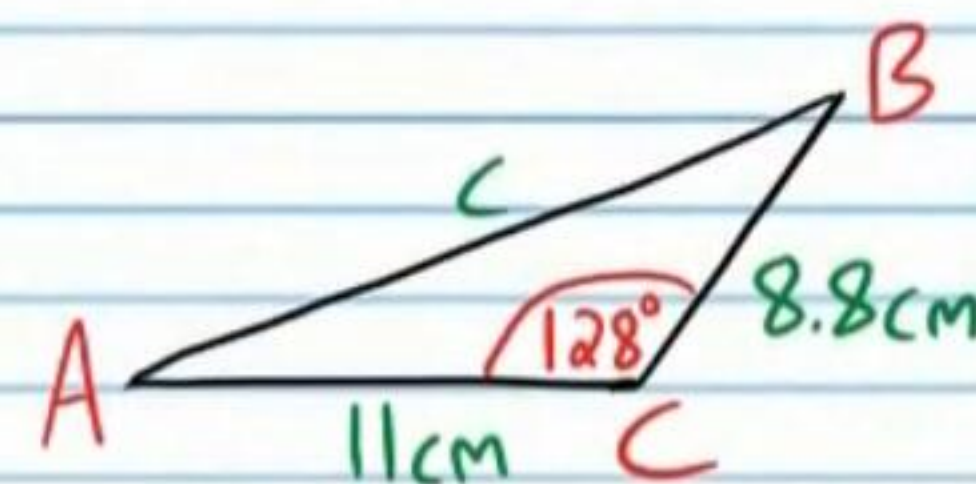
$$c^2 = 77.44 + 121 - 193.6 \cos 128^\circ$$

$$c^2 \approx 198.44 - 193.6(-0.61566...)$$

$$c^2 \approx 198.44 + 119.192...$$

$$c^2 \approx 317.632...$$

$$\boxed{c \approx 17.8222 \text{ cm}}$$



⑰ $m \angle A$

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\frac{\sin A}{8.8} \approx \frac{\sin 128^\circ}{17.8222...}$$

$$\sin A \approx 0.38909...$$

$$A \approx \sin^{-1}(0.38909...)$$

$$\boxed{A \approx 22.898^\circ}$$

⑱ $m \angle B$

$$A + B + C = 180^\circ$$

$$22.898^\circ + B + 128^\circ \approx 180^\circ$$

$$\boxed{B \approx 29.1^\circ}$$

① $C = 71^\circ$, $a = 5$ in, $c = 6$ in

I) Find $m\angle A$.

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\frac{\sin A}{5} = \frac{\sin 71^\circ}{6}$$

$$\sin A = \frac{5 \sin 71^\circ}{6}$$

$$\sin A \approx 0.78793...$$

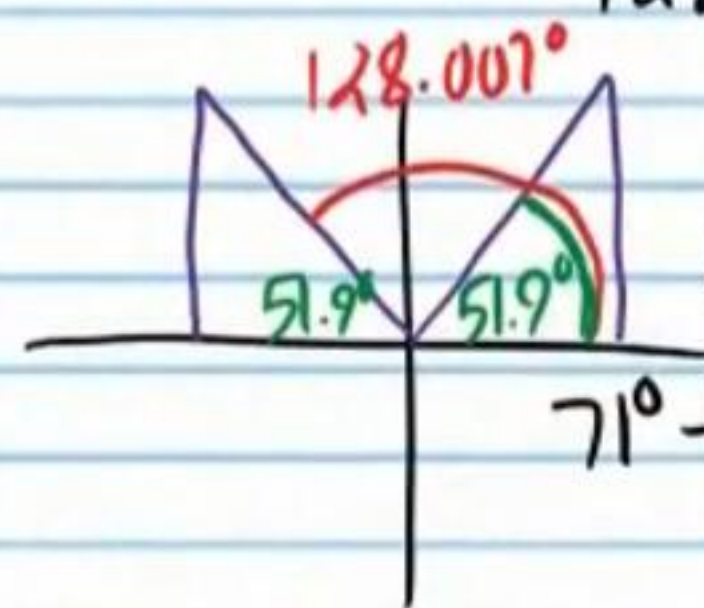
$$A \approx \sin^{-1}(0.78793...)$$

$$A \approx 51.9926^\circ$$

$$C + A < 180^\circ$$

$$71^\circ + 51.9926^\circ < 180^\circ$$

$$122.99^\circ < 180^\circ$$



One Triangle

$$71^\circ + 128.007^\circ < 180^\circ$$

$$199^\circ < 180^\circ$$

II) Find $m\angle B$

$$A + B + C = 180^\circ$$

$$51.9926^\circ + B + 71^\circ \approx 180^\circ$$

$$B \approx 57.0074^\circ$$

III) Find b .

$$\frac{\sin B}{b} = \frac{\sin C}{c}$$

$$\frac{\sin 57.0074^\circ}{b} \approx \frac{\sin 71^\circ}{6}$$

$$b \approx \frac{6 \sin 57.0074^\circ}{\sin 71^\circ}$$

$$b \approx 5.322 \text{ in}$$

②② $B = 22^\circ$, $b = 16$ km, $a = 51$ km

I) Find $m\angle A$.

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin A}{51} = \frac{\sin 22^\circ}{16}$$

$$\sin A = \frac{51 \sin 22^\circ}{16}$$

$$\sin A \approx 1.194$$

No Triangle Possible

②1) $A = 146^\circ$, $a = 26$ mi, $c = 30$ mi

I) Find $m\angle C$.

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\frac{\sin 146^\circ}{26} = \frac{\sin C}{30}$$

$$\frac{30 \sin 146^\circ}{26} = \sin C$$

$$\sin C \approx 0.64522\dots$$

$$C \approx 40.1823^\circ\dots$$

$$A + C < 180^\circ$$

$$146^\circ + 40.1823^\circ < 180^\circ$$

$$186.1823^\circ < 180^\circ \quad \times$$

No Triangle Possible

②② $C = 10^\circ$, $b = 18 \text{ mm}$, $c = 15 \text{ mm}$

I) Find $m\angle B$.

$$\frac{\sin B}{b} = \frac{\sin C}{c}$$

$$\frac{\sin B}{18} = \frac{\sin 10^\circ}{15}$$

$$\sin B = \frac{18 \sin 10^\circ}{15}$$

$$\sin B \approx 0.20837\dots$$

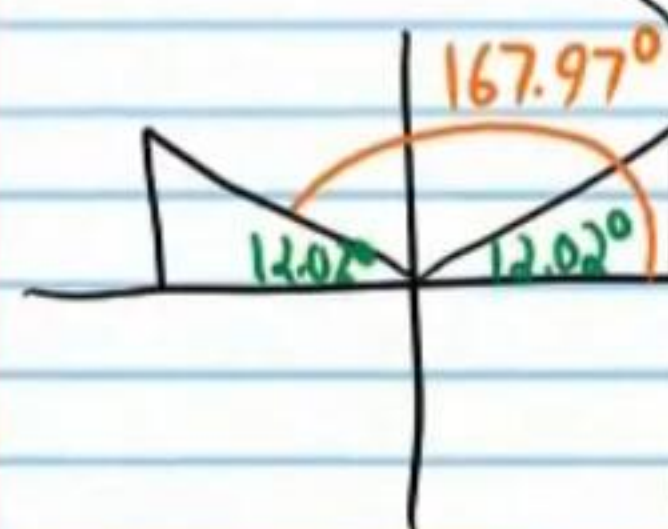
$$B \approx \sin^{-1}(0.20837\dots)$$

$$B \approx 12.0273^\circ$$

$$C + B_1 < 180^\circ$$

$$10^\circ + 12.02^\circ < 180^\circ$$

$$22.02 < 180^\circ \checkmark$$



$$C + B_2 < 180^\circ$$

$$10^\circ + 167.97^\circ < 180^\circ$$

$$177.97^\circ < 180^\circ \checkmark$$

II) Find $m\angle A$

$$A_1 + B_1 + C = 180^\circ$$

$$A_1 + 12.0273^\circ + 10^\circ \approx 180^\circ$$

$$A_1 \approx 157.9727^\circ$$

III) Find a .

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\frac{\sin 157.9727^\circ}{a} \approx \frac{\sin 10^\circ}{15}$$

$$a \approx \frac{15 \sin 157.9727^\circ}{\sin 10^\circ}$$

$$a \approx 32.4 \text{ mm}$$

$$I) B_2 \approx 167.97269^\circ$$

II) Find $m\angle A_2$.

$$A_2 + B_2 + C = 180^\circ$$

$$A_2 + 167.97269^\circ + 10^\circ \approx 180^\circ$$

$$A_2 \approx 2.0273^\circ$$

III) Find a_2 .

$$\frac{\sin A_2}{a_2} = \frac{\sin C}{c}$$

$$\frac{\sin 2.0273^\circ}{a_2} \approx \frac{\sin 10^\circ}{15}$$

$$a_2 \approx \frac{15 \sin 2.0273^\circ}{\sin 10^\circ}$$

$$a_2 \approx 3.0558 \text{ mm}$$

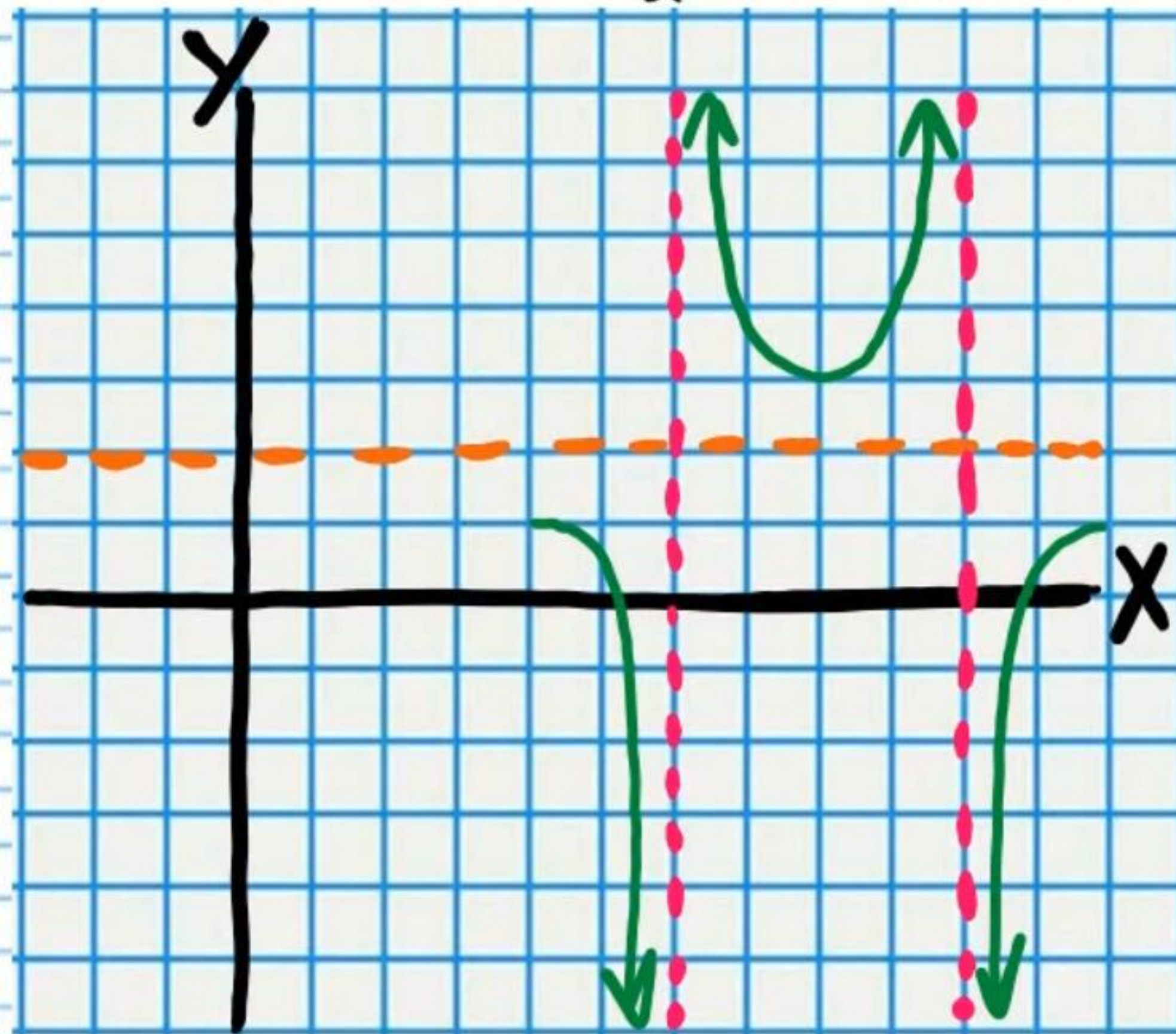
For the following problems, write the equation of the midline, and find the period, amplitude, and phase shift. Then, graph the equation. *Only one cycle of the graph is necessary.*

②③ $y = -\sec\left(\frac{x}{2} - 180^\circ\right) + 2$

Midline Equation: $y = 2$

Period: $P = \frac{360^\circ}{\omega} = \frac{360^\circ}{\frac{1}{2}} = 720^\circ$

Phase Shift: $PS = \frac{\Phi}{\omega} = \frac{180^\circ}{\frac{1}{2}} = 360^\circ$



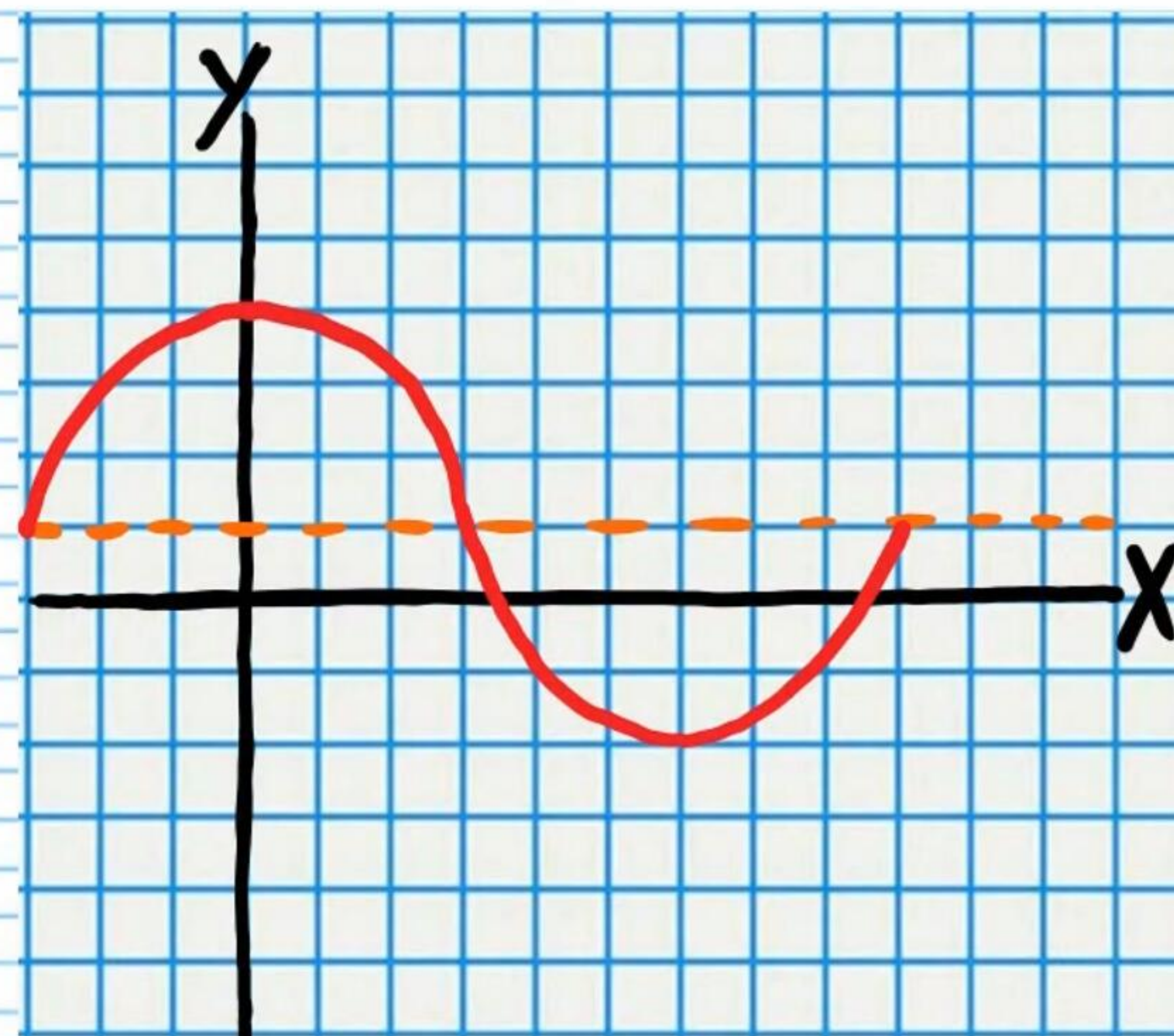
②④ $y = 3\sin\left(\frac{x}{3} + 90^\circ\right) + 1$

Midline Equation: $y = 1$

Period: $P = \frac{360^\circ}{\omega} = \frac{360^\circ}{\frac{1}{3}} = 1,080^\circ$

Phase Shift: $PS = \frac{\Phi}{\omega} = \frac{-90^\circ}{\frac{1}{3}} = -270^\circ$

Amplitude: $A = |3| = 3$



②⑤ $y = \cot(2x + 360^\circ) - 2$

Midline Equation: $y = -2$

Period: $P = \frac{180^\circ}{w} = \frac{180^\circ}{2} = 90^\circ$

Phase Shift: $PS = \frac{\Phi}{w} = \frac{-360^\circ}{2} = -180^\circ$

